

COA IMPORTANT QUESTIONS

UNIT - 1

1. Explain the functional units of digital system and their interconnections in detail.

Or

Explain the von neumann architecture of computer system in detail, Also give advantage and disadvantage of this architecture as compared to other architecture.

2. Draw the block diagram of computer CPU showing all the building basic block, and hence explain the same in detail. Also give suitable reason as how such arrangement can work as computer can work as a computer if connected to memory, I/O interface.
3. What is Bus? Draw the general structure of system bus and hence explain the different types of system bus with their relative functions.
4. What are the different bus structures available in computer system, explain in details with necessary block diagram required and also mention advantage and disadvantage of each architecture in detail.
5. What do you mean by term bus arbitration? Why we need bus arbitration? Explain different mechanism of bus arbitration in detail with required block diagram, also mention relative advantage and disadvantage of each mechanism.

Or

Compare and contrast different mechanism of bus arbitration in tabular format.

6. What do mean by RTL? How it is useful in computer system? Enlist different feature and component of RTL in detail.
7. What is memory transfer? What are the different registers associated for memory transfer? Discuss.
8. What do you mean by Bus transfer? Draw a diagram of bus system in which it uses:
- Three state buffers and a decoder.
 - MUX which has 4 register of size 4 bit each.
9. Explain general purpose register based organization in detail.
10. What is stack? Give the organization of register stack with all necessary elements and explain the working of PUSH and POP operations in detail.
11. What is memory stack? Compare register stack and memory stack. Explain the role of memory stack in managing subroutines.
12. Explain an accumulator based central processing unit organization with block diagram.
13. What do you mean by term processor organization? Explain different types of processor organization in detail.
14. What do you mean by term addressing mode? Why do we need addressing mode? How addressing mode is significant for referring memory? List and explain different types of addressing mode in detail.
15. Define the term control word with an example, also explain the general format of control word and hence explain the same.

UNIT - 2

1. With the help of block diagram explain sequential arithmetic and logical unit in detail and hence explain how we can implement and design such ALU unit.
2. Design a 4-bit carry look ahead adder and explain its operation with an example.

Or

Explain in detail the principle of carry look head adder and design 4-bit CLA adder.

3. With the help of flowchart explain booth's multiplication algorithm for 2's complement number, and hence multiply the following numbers using the same:

a. $17 * 15$

b. $-14 * 7$

c. $13 * -6$

d. $-12 * 9$

4. Explain Booth's algorithm with its hardware implementation.
5. Explain array multiplier method with the help of example.
6. With the help of flowchart explain restoring and non-restoring division operation, and hence divide the following number using both method restoring and non-restoring:
- a. $13 / 12$ b. $19 / 4$ c. $15 / 4$ d. $12 / 8$
7. Draw the data path of sequential n-bit binary divider.
8. How floating point numbers are represented in computer, also give IEEE 754 standard for 32-bit and 64-bit floating point number format, hence represent $(-1640.215)_{10}$ and $(1460.125)_{10}$ in single precision and double precision format.
9. Draw a flowchart for adding and subtracting two fixed point binary numbers where negative numbers are signed 1's complement presentation.

UNIT - 3

1. What is an instruction in the context of computer organization? Explain the purpose of the various elements of an instruction with the help of a instruction format.
2. Describe the types of instructions on the basis of address fields used in the instruction and hence evaluate the arithmetic statement $X = A * [B + C * D + E] / F * (G + H)$ using a general register computer with three address, two address, one address and zero address instruction format program to evaluate the expression.
3. Define instruction cycle and divide instruction cycle into sub cycles with the help of diagram, explain the sequence in which sub cycles are executed.
4. Write the steps in fetching a word from memory. Differentiate between a branch instruction and call subroutine instruction.
5. What are the different categories of micro-operations or instructions that may be carried out by CPU? Explain each category of micro-operations giving one example for each.
6. What is CISC? Explain its characteristics.
7. What is RISC? Explain its various characteristics.
8. Differentiate among the following:
- RISC and CISC based microprocessor.
 - Hardwired and Micro-programmed control unit
 - Horizontal organization and Vertical organization (Note: These are technique to group control signal)

9. What do you understand by hardwired control unit? With the help of Block diagram explain the working of hardwired control unit in detail with relative advantages and disadvantages, also explain various methods to design hardwired control unit in detail.
10. What do you understand by micro-programmed control unit? With the help of Block diagram explain the working of micro-programmed control unit in detail with relative advantages and disadvantages.

Or

With the help of necessary diagram explain the two technique of generating control signals.

11. What is a micro-program sequencer? With block diagram, explain the working of micro-program sequencer.

Or

With the help of block diagram explain address selection for control memory.

12. Explain the concept of vertical and horizontal multi-programming in detail.
13. Write Short notes on the following:
 - a. Micro-operation
 - b. Micro-code
 - c. Micro-program
 - d. Control memory
 - e. Wide branch addressing
 - f. Pre – fetching micro instruction
 - g. Micro – instruction
14. What do you mean by term pipelining and how the idea of pipelining used in a computer? Explain different types of pipelining, and also define different modes of pipelining available.
15. What are the various pipeline performance measures? Explain in detail.

UNIT – 4

1. Explain concept of memory and hence describe memory hierarchy in detail.
2. Explain semiconductor RAM. Enlist the types of semiconductor memory and hence give e structure of commercial 8M × 8 bit DRAM chip.

Or

Explain dynamic RAM and static RAM and hence give the structure of commercial 8M×8 bit DRAM chip.

3. Explain 2D, 2.5D memory organization.
4. What is ROM? Explain the types of semiconductor based ROM memories.
5. How main memory is useful in computer system? Explain the memory address map of RAM and ROM.
6. Write short note on cache memory. Discuss the design issues in cache design and hence explain how does it affect the performance of the computer system and how the performance of cache memory is measured?
7. What is meant by cache mapping? What are different types of mapping? Discuss different mapping techniques with examples.
8. Explain the different method of writing into cache in detail.
9. Explain the term locality of reference and hence differentiate among spatial locality and temporal locality.
10. Explain various page replacement algorithms in detail with example, hence explain the concept of belady's anomaly and also determine which page replacement algorithm suffer from this problem.
11. Explain auxiliary memory. What are the commonly used auxiliary memories?
12. Explain the concept of virtual memory in detail, also determine the methods to implement virtual memory.
13. What do you mean by CAM? Explain its major characteristics.

14. Explain the following memory schemes, also discuss their need and working :
 - a. Interleaved memory
 - b. Associative memory

15. Discuss the conceptual organization of a multilevel memory system used in computers.

UNIT – 5

16. What do you mean by the term I/O Interface? Enlist different function of I/O interface in detail. Also give suitable reason for the need of I/O interface in transferring information between internal storage and external I/O devices.

17. Write short notes on :- ① Peripheral Devices, ② I/O bus and ③ I/O Command.

18. What do you mean by I/O processor? How I/O processor communicate with CPU? Justify your answer with necessary block diagram required.

19. Explain the term interrupt in detail with it's classification, also enlist the sequence that take place when an interrupt occur.

20. When a device interrupt occurs how does the processor determine which device has issued the interrupt, hence determine how system resolve the priority of interrupt?

21. With the help of flow diagram explain programmed I/O method and interrupt initiated I/O method for controlling input-output operation in detail.

22. Explain DMA based data transfer method in detail, hence explain how DMA is able to transfer data without intervention of CPU.

23. Differentiate among the following:

- a. Isolated I/O & Memory Mapped I/O
- b. Vectored interrupt & Non-Vectored interrupt
- c. Serial transmission & Parallel transmission
- d. Synchronous data transfer & Asynchronous data transfer

24. Write short notes on the following:

- a. Input Output(I/O) Processor
- b. Input Output(I/O) Port
- c. Input Output(I/O) Channel

25. What do you mean by asynchronous data transfer? With the help of block diagram and timeline diagram explain different method of asynchronous data transfer in detail.

26. What do you mean by serial communication? Explain different mode of serial communication in detail.

27. Answer the following as asked:

- a. Why Read and Write control line in a DMA Controller are is Bi-directional?
- b. Describe cycle stealing in DMA
- c. What is the use of modem in synchronous communication?